

Restaurant Ordering System

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Python Restaurant POS System

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Problem Statement

- ▶ The Point of Sales (POS) system solves core business problems like slow, error-prone manual transactions, poor inventory tracking, and disorganized sales data by automating payments, managing stock in real-time, and centralizing customer/sales info, leading to faster service, fewer mistakes, better decision-making, and improved customer satisfaction.
- ▶ It is important because a POS system helps businesses streamline operations, proper management, and shows vital information for both the customer and the business owners.
- ▶ Who are the intended users? The intended are business owners who want to boost their business and have proper management

Project Objectives

1. The first goal of the project is to make an entry-level POS system using Python fundamentals we have learn
2. The second goal of the project is to make a POS system that is accessible to the admin using a password system to check and add items, pieces to the menu. Then for the customer to use the POS for ordering and see the total bill at the end.

System Overview / Concept

The system works in two ways. First, the admin can enter the system using a password and a login method in which the admin user can check the menu, add or remove items from the menu, and manage orders. Second, the guest can enter their name and add items like appetizer, main course, drinks, and dessert, which also shows the price, which finish it will print out a table with the list of items and their total at the end.

Key Python Features Used

- ▶ [] Conditional Statements (obviously needed)
- ▶ [] Loops (also need to order multiple items)
- ▶ [] Function (different functions for different actions)
- ▶ [] List, or Dictionary (for storage of food items, and customer orders)
- ▶ [] Read/Write/Update/Delete to File (updating menus, and storage in a JSON file)
- ▶ [] Interaction with the user through Terminal or other libraries (interaction through the terminal, with iterfzf acting as a search bar)

Core Features / Functions

- ▶ *show the file structure
- ▶ **show the main functions*
- ▶ *show the json file

Demo

Go to `restaurant.k1mch1.space`

Challenges During Implementation

- ▶ Git version control
- ▶ deciding on the correct design decisions
- ▶ deciding on the correct way to implement a feature
- ▶ naming variables

Future Enhancements

- ▶ Get it on the WEB! Use REST api... man...
- ▶ Get a GUI or a TUI
- ▶ Add transactions!
- ▶ Enable users to update their orders or sth...
- ▶ Improve security!
- ▶ Write cleaner, maintainable code with better documentation
- ▶ Get a database instead of storing in a damn JSON
- ▶ Backup said database too!

Conclusion

What we learned:

- ▶ all the basic features of a programming language
- ▶ data representation and storage
- ▶ a few functions from a lot of libraries
- ▶ a bit of git and version control
- ▶ a bit of hashing
- ▶ documenting code, and writing maintainable code

Member Contributions

- ▶ commit history
- ▶ contributor's page
- ▶ todo at readme

Thanks For Listening! Any questions?